TEACHER GUIDE · GRADE 2 · EVENTS & PARALLEL SCRIPTS

Debug the Events

Quick facilitation notes & answer key

Goal

The challenge: students DEBUG an event bug (C3.2 read/alter + C3.1 write). Pixel never moves because its receive hat ('when I receive jump') does not match Bit's broadcast ('go'), so the event never fires. Students diagnose the message mismatch and fix it by making the two messages the same. Debugging a broken event link is the capstone of this subject.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: two 'message' cards reading 'go' and 'jump'; hold them up to show they do not match, then swap one so they do.

Run it (about 20 min)

1. I DO: run it — 'Bit goes 0 forward 2 → 2 and shouts go. But Pixel is waiting for jump, so it never hears its message.'
2. WE DO: compare the two message words — 'go' vs 'jump'. They are different, so Pixel's hat never fires.
3. YOU DO: students fix one message so both say the same word, then run Pixel: 5 back 2 → 3.
4. Connect: 'A receive hat needs the exact message it is waiting for — check the words match.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 0 → forward 2 → 2. Pixel does NOT move (circle NO); it stays on its home number 5.

Ex 2 — Bit broadcasts 'go'; Pixel waits for 'jump'. The messages do not match, so Pixel's receive event never fires and it never starts.

Ex 3 — fix: change Pixel's hat to 'when I receive go' (or change Bit's broadcast to 'jump'). With matching messages, Pixel runs: 5 → back 2 → 3.

Fix note: either edit works as long as BOTH messages end up the same word; then Bit 2 and Pixel 3.

Watch for & differentiate

Common error: changing a move block instead of a message — the bug is the EVENT (the message), not Pixel's motion.

Simplify: tell students to circle the two message words and check if they are the same.

Extend: ask what happens if Bit's 'broadcast go' block is deleted entirely (Pixel never starts no matter what its hat says).

Success indicator: the child reports Pixel stuck at 5 before, names the message mismatch, and fixes one message so Pixel lands on 3.